CprE 3810: Computer Organization and Assembly Level Programming

Team Contract – Project Part 2

Project Teams Group #: C_01

Team Members: Graeme Anderson

Colten Stevens

Discuss the following aspects of teamwork with your team – make sure to get input from each member. Write down your team's consensus for each of the bolded headings. Italicized text contains instructions and examples and should be deleted once you've read it. Please see the example contract for rough length expectations.

Course Goals: *List and acknowledge the goals of your individual team members. Examples may include:*

- *learn everything about computer architecture*
- *know enough to understand security risks posed by hardware primitives*
- *get an A/B/C/Pass in the course*
- *minimize the number of lost points*
- *prepare myself for a career in hardware design*
- *prepare myself to be able to do research involving FPGAs*
- *be able to explain the workings of a stored-program computer from gates to C*

Course Goals:

- ***get an A in the course***
- ***fully understand every possible instruction executed in out processor***
- ***create processor with data hazard, full forwarding and register bypass capabilities***
- ***have out synthesis of our pipelined register be 1.5 times faster than non-pipelined***

Team Expectations:

- **Conduct:** *What are the expectations for personal conduct of group members? Group members must show up to all agreed upon meeting times or notify why missing and do the work you say your are going to do by agreed upon date.*
- **Communication:** *What is the best mode of communication for the group? How often should communication occur? How fast should a response be expected? The best mode of communication has been snap chat texting this communication should happen around 5 times a week with checkups and be answerd within the day*

- **Group conventions:** *Naming conventions? Compilation and simulation methodology? Testbench strategies? Do file usage? Version control strategies? Commenting standards?*
  If a part of the processor is implemented as a black box that black box should be test benched before it is integrated. Naming conventions should be very clear so it can be understood by all group member what a signal is for.
- **Meetings:** *Given the significant portion of the course that the lab covers, it is expected that your team will spend more time working on the labs than in your scheduled lab sections. How will your group expect to handle this? Please include at least two additional times outside of lab that your team can meet (preferably in-person).*
  Meet in person every Sunday nigh and Tuesday night. Work outside of lab individually when applicable
- **Peer Evaluation Criteria:** *Now that you have experience working on a 381 lab with a team, please create a brief criteria for how effort and contribution are defined. Note that teams with **vastly** divergent scores may require a meeting with course instructor and result in different grades for different group members. Teams with reasonably equitable scores will receive the same grade.*
  Effort can be measured in different ways based on overall understanding. Same amount of man hours should be applied by group members

Role Responsibilities: *Complete the following planning table. Each lab part should be the responsibility of one team member. Also make sure that no one team member is the lead on both the design and test aspects of a single lab part. These guidelines aid in all students having a complete view of the lab. Note that the non-lead is encouraged to participate and support the lead wherever possible, increasing both the quality of the lab part and each team member's knowledge.*

Lab Part		Estimated Time	Design		Test	
			Lead	Deadline	Lead	Deadline
Software-Scheduled Pipeline	Control Signals	0.5 hr	G	11/5	G	11/5
	Datapath	3 hr	C	11/5	C	11/5
	Testing	3 hr	G	11/5	G	11/5
	Synthesis (human effort)	0.5 hr	C	11/5	C	11/5
Hardware-Scheduled Pipeline	Pipeline Register Update	1 hr	C	11/12	C	11/12
	Data Hazard Avoidance	4 hr	G	11/12	G	11/12
	Control Hazard Avoidance	2-6 hr based on group size	C	11/19	C	11/19
	Integration (Hardware-Schedule Pipeline)	3 hr	G	11/31	G	11/31
	Testing	3 hr	C	11/31	C	11/31
	Synthesis	0.5 hr	G	12/1	G	12/1

*Estimated Time is given as a **very rough** guide for even distribution of tasks assuming you've already read through the lab document and have the prerequisite knowledge. Please note that to be done properly, the test programs will require significant time investment, but will result in a much stronger final design.*

Integrity of Work: *Do not delete the following.* We agree that the work we provide to other team members and ultimately submit for a grade is a direct result of our own work as described in the course syllabus. Specifically, we will generate all VHDL code ourselves and not copy VHDL code from online sources, other groups, book companion material, or past student projects to which anyone outside of my team has contributed.

Student Signature Graeme Anderson **Date** 10/29/2025

Student Signature Colten Stevens **Date** 10/29/2025